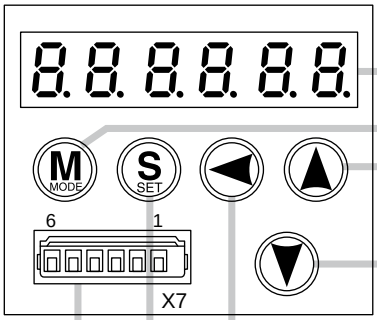
Setup with the Front Panel



Display LED (6-digit)
Switch to error display screen when error occurs, and LED will flash (about 2Hz).
LED will flash slowly (about 1Hz) when warning occurs.

Mode switching button
(valid at SELECTION display)
Press this to switch 4 kinds of mode.

1) Monitor Mode
2) Parameter Set up Mode
3) EEPROM Write Mode
4) Auxiliary Function Mode

SET Button (valid at any time)
Press this to switch SELECTION and EXECUTION display.

Press these to change display and data, select parameters and execute actions.
(Change/Selection/Execution is valid to the digit which decimal point flashes.)
Numerical value increases by pressing, ▲, decreases by pressing ▼.

Shifting of the digit for data changing to higher digit.

X7
Output connector for monitor

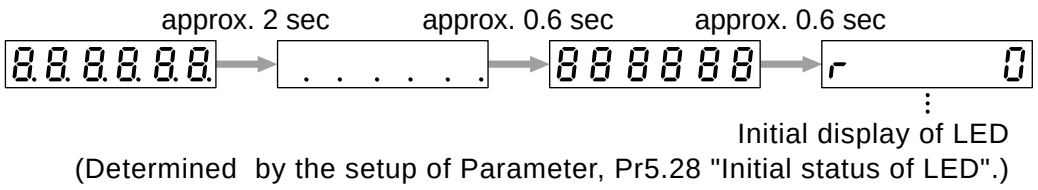
Note For connector X7, refer to P.2-27 “Wiring to the connector, X7”.

Caution

Initial Status of the Front Panel Display (7 Segment LED)

Status

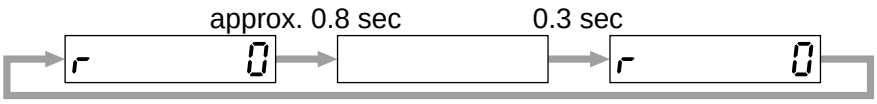
Front panel display shows the following after turning on the power of the driver.



Initial display of LED
(Determined by the setup of Parameter, Pr5.28 "Initial status of LED".)

Upon Occurrence of an Alarm

If a driver alarm is generated, the front panel display shows the following repeatedly.



Repeatedly(during continuous alarm)

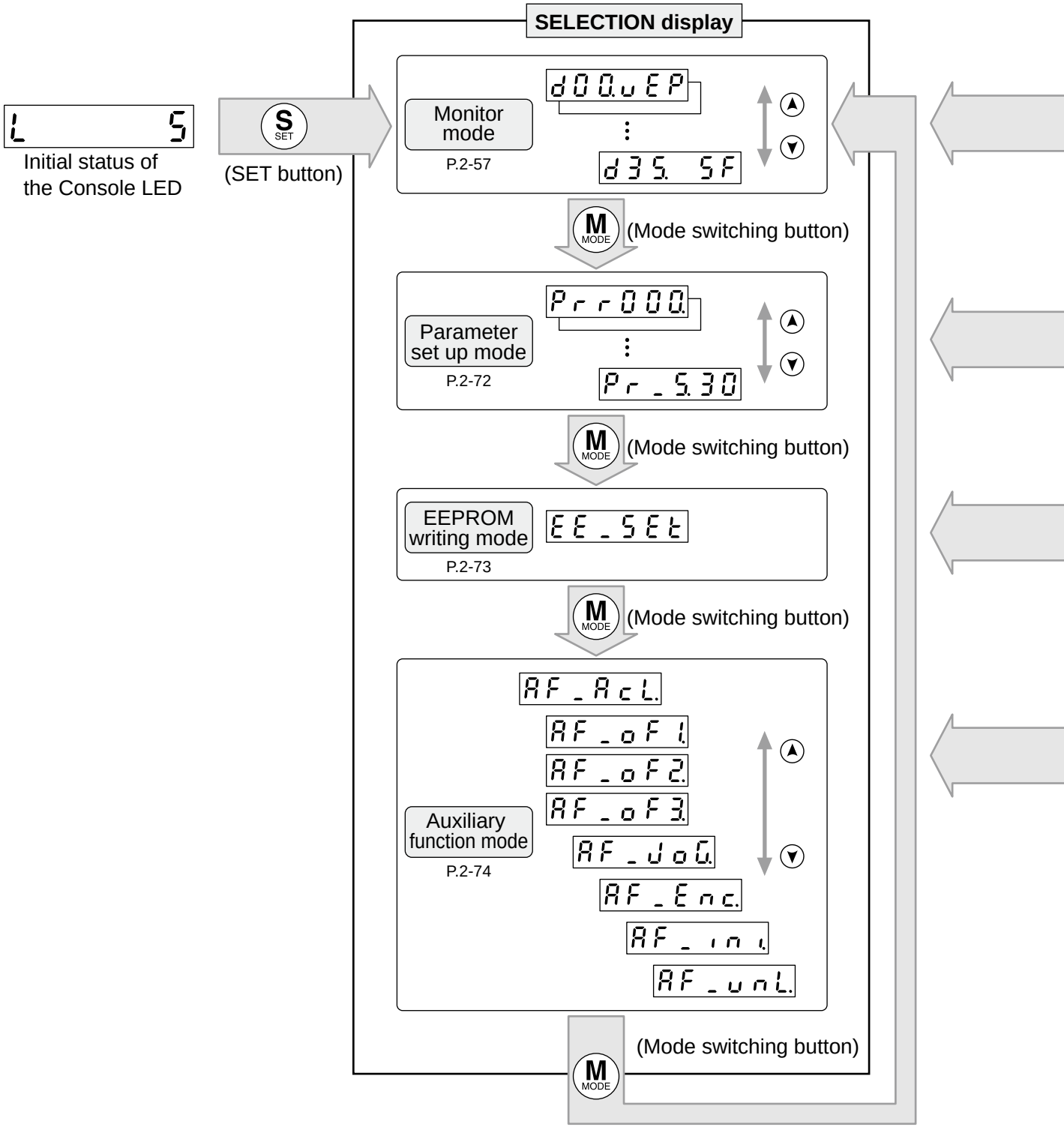
Below shows possible cause of an alarm.

alarm No.	Alarm	Content
A0	Overload protection	Load factor is 85% or more the protection level.
A1	Over-regeneration alarm	Regenerative load factor is 85% or more the protection level.
A2	Battery alarm	Battery voltage is 3.2 V or lower.
A3	Fan alarm	Fan has stopped for 1 sec.
A4	Encoder communication alarm	The number of successive encoder communication errors exceeds the specified value.
A5	Encoder overheat alarm	The encoder detects overheat alarm.
A6	Oscillation detection alarm	Oscillation or vibration is detected.
A7	Lifetime detection alarm	The life expectancy of capacity or fan becomes shorter than the specified time.
A8	External scale error alarm	The feedback scale detects the alarm.
A9	External scale communication alarm	The number of successive feedback scale communication errors exceeds the specified value.

15. How to Use the Front Panel

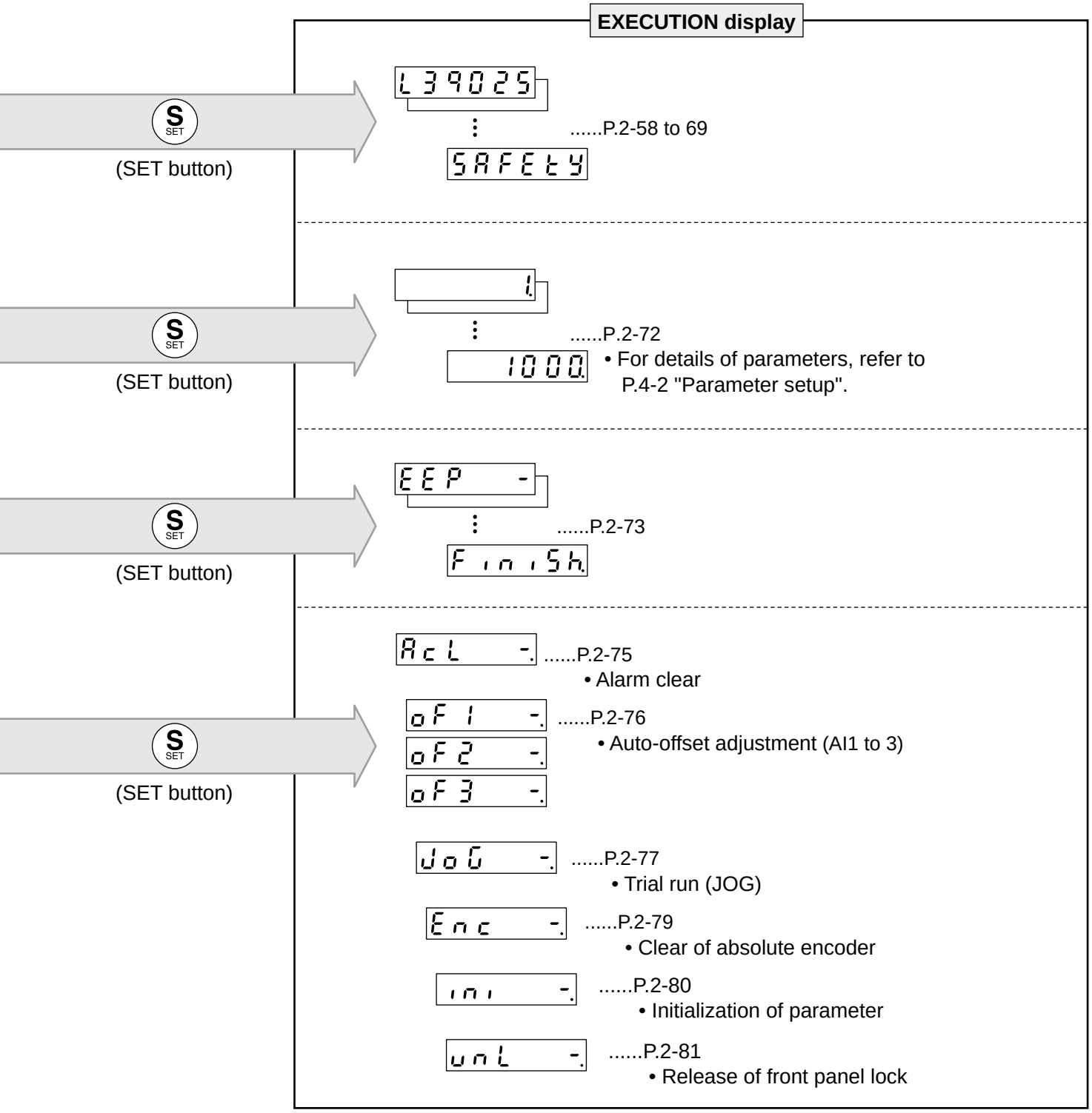
Structure of Each Mode

Use each button on the touch panel to select the structure and switch the mode.



Note You can change the flashing decimal point with ◀, then shift the digit for data change "."

Caution On power-up, the monitor mode executed is displayed according to the setup of Pr5.28 LED initial status.


EXECUTION display
L 39025
SAFEty
.....P.2-58 to 69
(SET button)
1.
1000.
.....P.2-72
• For details of parameters, refer to P.4-2 "Parameter setup".
(SET button)
EEP -
FinISh.
.....P.2-73
(SET button)
AcL -.
.....P.2-75
• Alarm clear
oF 1 -.
oF 2 -.
oF 3 -.
.....P.2-76
• Auto-offset adjustment (AI1 to 3)
(SET button)
JoG -.
.....P.2-77
• Trial run (JOG)
Enc -.
.....P.2-79
• Clear of absolute encoder
ini -.
.....P.2-80
• Initialization of parameter
unL -.
.....P.2-81
• Release of front panel lock

15. How to Use the Front Panel

Setup of front panel lock

Outline

To prevent operational error e.g. unintentional parameter modification, the front panel may be locked.
Once locked, operations on the panel are limited as follows:

Mode	Locked panel conditions
Monitor Mode	No limitation: all monitored data can be checked.
Parameter Set up Mode	No parameter can be changed but setting can be checked.
EEPROM Write Mode	Cannot be run. (No display)
Auxiliary Function Mode	Cannot be run except for “Release of front panel lock”. (No display)

How to operate

• Related parameters

Parameter No.		Title	Function
Class	No.		
5	35	Setup of front panel lock	Locks the operation attempted from the front panel.

Lock and unlock can be made in one of two ways.

Procedure	Front panel	Setup support software PANATERM
Lock	(1) Set Pr5.35 “Front panel lock” to 1, and writ the setting to EEPROM. (2) Turn on power to the driver. (3) The front panel is locked.	
Unlock	(1) Execute the auxiliary function mode, front panel lock release function. (2) Turn on power to the driver. (3) The front panel is unlocked.	(1) Set Pr5.35 “Front panel lock” to 0, and writ the setting to EEPROM. (2) Turn on power to the driver. (3) The front panel is unlocked.

Note For details of front panel lock release, refer to P.2-81.

15. How to Use the Front Panel

Monitor Mode (SELECTION display)

To change the monitor display setting, select the display option to be changed from " **SELECTION** display", and press (S) to change to " **EXECUTION** display". After completion of changing, press (S) to return to the selection display,

Monitor Mode SELECTION display

Display example	Description	Pages to refer
d00.uEP	Positional command deviation	P.2-58 (1)
d01.SPd	Motor speed	P.2-58 (2)
d02.cSP	Positional command speed	P.2-58 (2)
d03.cUL	Velocity control command	P.2-58 (2)
d04.tr9	Torque command	P.2-58 (2)
d05.nPS	Feedback pulse sum	P.2-59 (3)
d06.cPS	Command pulse sum	P.2-59 (3)
d08.FPS	External scale feedback pulse sum	P.2-59 (3)
d09.cnt	Control mode	P.2-59 (4)
d10. io	I/O signal status	P.2-60 (5)
d11.Ain	Analog input value	P.2-61 (6)
d12.Err	Error factor and reference of history	P.2-62 (7)
d13. rn	Alarm Display	P.2-64 (8)
d14. rG	Regenerative load factor	P.2-65 (9)
d15. oL	Over-load factor	P.2-65 (9)
d16.Jrt	Inertia ratio	P.2-65 (9)
d17. ch	Factor of no-motor running	P.2-66 (10)

Display example	Description	Pages to refer
d18. ict	No. of changes in I/O signals	P.2-67 (11)
d20.Abs	Absolute encoder data	P.2-67 (12)
d21.AES	Absolute external scale position	P.2-68 (13)
d22.rEc	No. of encoder/ external scale communication errors monitor	P.2-68 (14)
d23. id	Communication axis address	P.2-68 (15)
d24.PEP	Encoder positional deviation [Encoder unit]	P.2-68 (16)
d25.PFE	External scale deviation [External scale unit]	P.2-69 (17)
d26.hYb	Hybrid deviation [Command unit]	P.2-69 (18)
d27. Pn	Voltage across PN [V]	P.2-69 (19)
d28. no	Software version	P.2-69 (20)
d29.ASE	Driver serial number	P.2-69 (21)
d30.MSE	Motor serial number	P.2-69 (22)
d31. tE	Accumulated operation time	P.2-70 (23)
d32.Aud	Automatic motor recognizing function	P.2-70 (24)
d33.Ath	Driver temperature	P.2-70 (25)
d35. SF	Safety condition monitor	P.2-71 (26)

<Return to "Positional command deviation".>

EXECUTION display ↔ (S) (SET button)

Display shifts toward the arrowed direction by pressing (▲) and reversed direction by pressing (▼).

(M) (Mode switch button)

Parameter Setup Mode SELECTION display

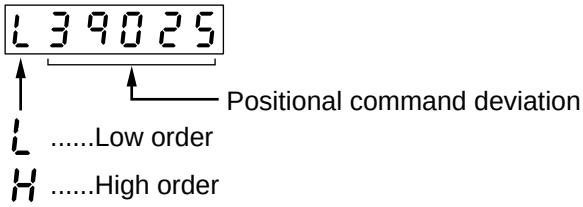
Note When you turn on the Product for the first time, display shows r 0. (at motor stall) To change this display, change the setup of Pr5.28 (Initial status of LED).

15. How to Use the Front Panel

Monitor Mode (EXECUTION display)

(1) Display of positional command deviation [command unit]

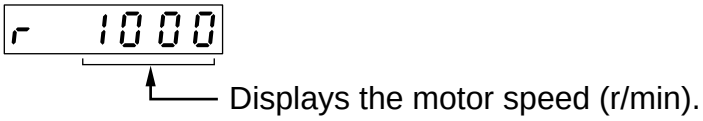
Displays positional deviation of the command unit in High order or Low order.



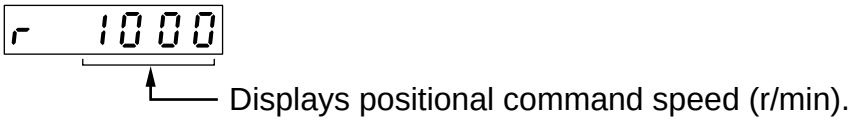
• To switch between Low order (L) and High order (H), press ◄.

(2) Display of motor speed, positional command speed, velocity control command and torque command

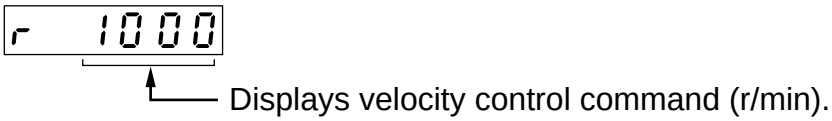
• Motor speed (r/min)



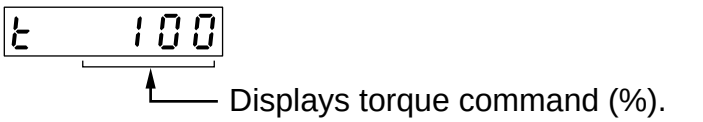
• Positional command speed (r/min)



• Velocity control command (r/min)

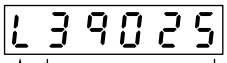


• Torque command (%)



(3) Display of Feedback Pulse Sum, Command Pulse Sum and External Scale Feedback Pulse Sum

• Feedback Pulse Sum [Encoder pulse]



Feedback Pulse Sum

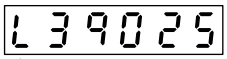
L .....Low order

H .....High order

• To switch between Low order (L) and High order (H), press ◄.



• Command Pulse Sum [Command Pulse]



Command Pulse Sum

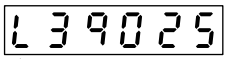
L .....Low order

H .....High order

• To switch between Low order (L) and High order (H), press ◄.



• External Scale Feedback Pulse Sum



External Scale Feedback Pulse Sum

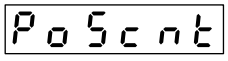
L .....Low order

H .....High order

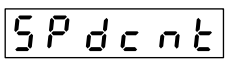
• To switch between Low order (L) and High order (H), press ◄.



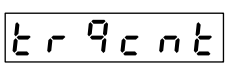
(4) Display of Control Mode



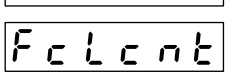
.....Position control mode



.....Velocity control mode



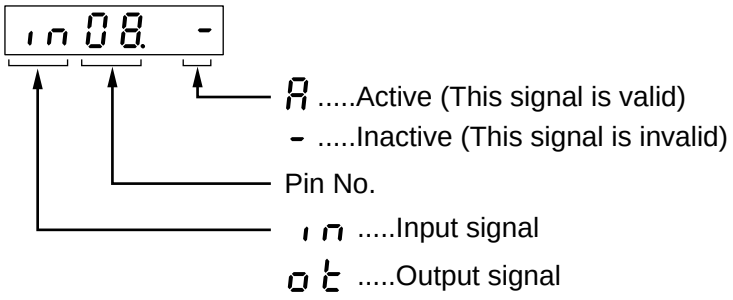
.....Torque control mode



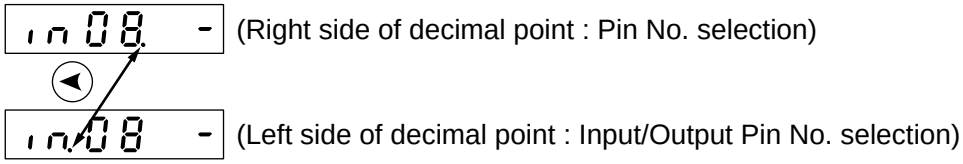
.....Full-closed control mode

(5) Display of I/O Signal Status

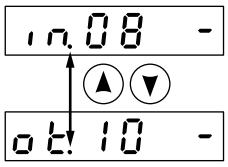
Displays the control input and output signal to be connected to connector X4.
Use this function to check if the wiring is correct or not.



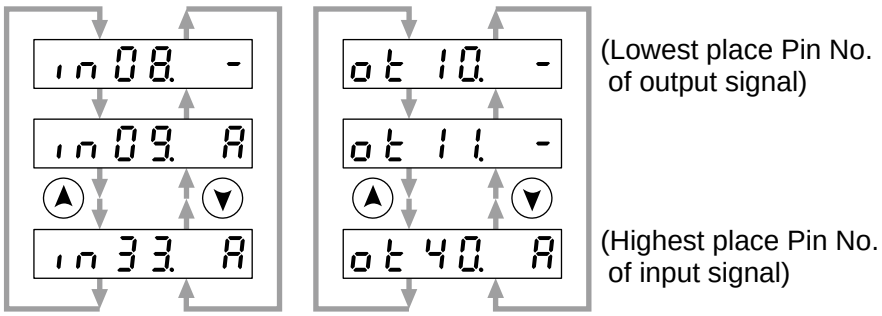
• Shift the flashing decimal point with ◀.



• Select In or Out by pressing ▲ or ▼ button.

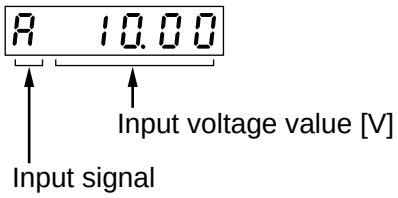


• Select the Pin No. to be monitored by pressing ▲▼.

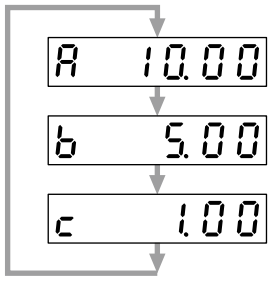


Note For detail of input/output signal, refer to P3-30 “Inputs and outputs on connector X4”
For detail of Error Code, refer to P.6-3 “Protective Function”.

(6) Display of Analog Input Value



•Select the signal No. to be monitored by pressings ▲ ▼.



(Analog input 1 value, unit [V])

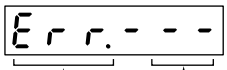
(Analog input 2 value, unit [V])

(Analog input 3 value, unit [V])

Displays the value after offset correction.

Caution Voltage exceeding ± 10V can not be displayed correctly.

(7) Display of Error Factor and Reference of History



Error code No. (- - - appears if no error occurs)

Err. .....Present error

E-0 .....History 0 (latest error)

E 13 .....History 13 (oldest error)

• You can refer the last 14 error factors (including present one)
Press ▲ ▼ to select the factor to be referred.

<List of error code No.>

Error code		Protective function	Attribute		
Main	Sub		History	Can be cleared	Immediate stop
11	0	Control power supply under- voltage protection		○	
12	0	Over-voltage protection	○	○	
13	0	Main power supply under-voltage protection (between P to N)		○	
	1	Main power supply under-voltage protection (AC interception detection)		○	
14	0	Over-current protection	○		
	1	IPM error protection	○		
15	0	Over-heat protection	○		○
16	0	Over-load protection	○	○*1	
18	0	Over-regeneration load protection	○		○
	1	Over-regeneration Tr error protection	○		
21	0	Encoder communication disconnect error protection	○		
	1	Encoder communication error protection	○		
23	0	Encoder communication data error protection	○		
24	0	Position deviation excess protection	○	○	○
	1	Velocity deviation excess protection	○	○	○
25	0	Hybrid deviation excess error protection	○		○
26	0	Over-speed protection	○	○	○
	1	2nd over-speed protection	○	○	
27	0	Command pulse input frequency error protection	○	○	○
	2	Command pulse multiplier error protection	○	○	○
28	0	Limit of pulse replay error protection	○	○	○
29	0	Deviation counter overflow protection	○	○	
30	0	Safety detection		○	
33	0	IF overlaps allocation error 1 protection	○		
	1	IF overlaps allocation error 2 protection	○		
	2	IF input function number error 1 protection	○		
	3	IF input function number error 2 protection	○		
	4	IF output function number error 1 protection	○		
	5	IF output function number error 2 protection	○		
	6	CL fitting error protection	○		
	7	INH fitting error protection	○		

Error code		Protective function	Attribute		
Main	Sub		History	Can be cleared	Immediate stop
34	0	Software limit protection	○	○	
36	0 to 2	EEPROM parameter error protection			
37	0 to 2	EEPROM check code error protection			
38	0	Over-travel inhibit input protection		○	
39	0	Analog input1 excess protection	○	○	○
	1	Analog input2 excess protection	○	○	○
	2	Analog input3 excess protection	○	○	○
40	0	Absolute system down error protection	○	○	
41	0	Absolute counter over error protection	○		
42	0	Absolute over-speed error protection	○	○	
43	0	Initialization failure	○		
44	0	Absolute single turn counter error protection	○		
45	0	Absolute multi-turn counter error protection	○		
47	0	Absolute status error protection	○		
48	0	Encoder Z-phase error protection	○		
49	0	Encoder CS signal error protection	○		
50	0	External scale connection error protection	○		
	1	External scale communication error protection	○		
51	0	External scale status 0 error protection	○		
	1	External scale status 1 error protection	○		
	2	External scale status 2 error protection	○		
	3	External scale status 3 error protection	○		
	4	External scale status 4 error protection	○		
	5	External scale status 5 error protection	○		
55	0	A-phase connection error protection	○		
	1	B-phase connection error protection	○		
	2	Z-phase connection error protection	○		
87	0	Compulsory alarm input protection		○	
95	0 to 4	Motor automatic recognition error protection			
Other number		Other error	○		

Note

History...The error will be stored in the error history.

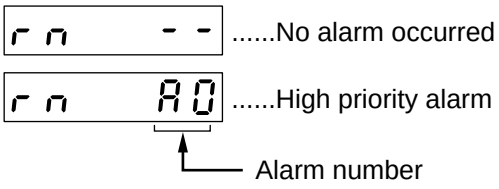
Can be cleared...To cancel the error, use the alarm clear input (A-CLR).
If the alarm clear input is not effective, turn off power, remove the cause of the error and then turn on power again.

Immediate stop...Instantaneous controlled stop upon occurrence of an error.
(Setting of “Pr.5.10 Sequence at alarm” is also required.)

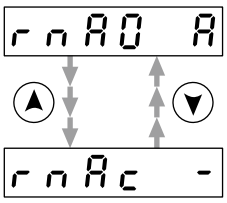
Caution

1) Certain alarms are not included in the history. For detailed information on alarms e.g. alarm numbers, refer to P.6-3.
2) When one of the errors which are listed in error history occurs, this error and history o shows the same error No.

(8) Alarm Display



• To display the alarm occurrence condition, press ▲ or ▼ button.

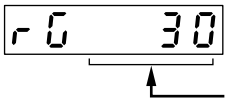


alarm No.	Alarm	Content	Latched time *1
A0	Overload protection	Load factor is 85% or more the protection level.	1 to 10s or ∞
A1	Over-regeneration alarm	Regenerative load factor is 85% or more the protection level.	10s or ∞
A2	Battery alarm	Battery voltage is 3.2 V or lower.	Fixed at ∞
A3	Fan alarm	Fan has stopped for 1 sec.	1 to 10s or ∞
A4	Encoder communication alarm	The number of successive encoder communication errors exceeds the specified value.	1 to 10s or ∞
A5	Encoder overheat alarm	The encoder detects overheat alarm.	1 to 10s or ∞
A6	Oscillation detection alarm	Oscillation or vibration is detected.	1 to 10s or ∞
A7	Lifetime detection alarm	Life expectancy of capacitor or fan is short.	Fixed at ∞
A8	External scale error alarm	The feedback scale detects the alarm.	1 to 10s or ∞
A9	External scale communication alarm	The number of successive feedback scale communication errors exceeds the specified value.	1 to 10s or ∞

*1 Alarms can be cleared by using the alarm clear. Because the all existing alarms are kept cleared while the alarm clear input (A-CLR) is kept ON, be sure to turn it OFF during normal operation. Either 1-10s or ∞ can be selected by using user parameter.
Exception: Battery alarm is fixed at ∞ because it is latched by the encoder.
Because the end of life alarm means that the life expectancy cannot be extended, the alarm is set at ∞.

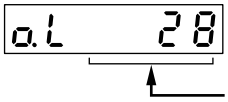
(9) Display of Regenerative Load Factor, Over-load Factor and Inertia Ratio

• **Regenerative Load Factor**



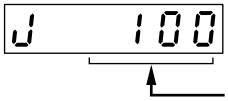
Display the ratio (%) against the alarm trigger level of regenerative protection.
This is valid when Pr0.16 (External regenerative resistor setup) is 0 or 1.

• **Over-load Factor**



Displays the ratio (%) against the rated load.
Refer to P.6-13, "Overload Protection Time Characteristics" of When in Trouble.

• **Inertia Ratio**

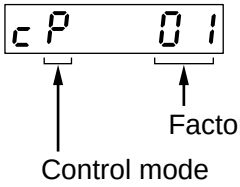


Displays the inertia ratio (%) .
Value of Pr0.04 (Inertia Ratio) will be displayed as it is.

Note For alarm function, refer to P.4-41"Pr4.40, Pr4.41".

10) Display of the Factor of No-Motor Running

Displays the factor of no-motor running in number.



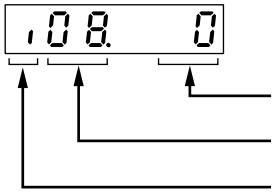
P .....Position control
t .....Torque control
S ..... Velocity control
F ..... Full-closed control

• Explanation of factor No.

Factor No.	Factor	Related Control Mode				Content
		P	S	T	F	
flashing	Occurrence of error/alarm	P	S	T	F	An error is occurring, and an alarm is triggered.
00	No particular factor	P	S	T	F	No factor is detected for No-motor run. The motor runs in normal case.
01	Main power shutoff	P	S	T	F	The main power of the driver is not turned on.
02	No entry of SRV-ON input	P	S	T	F	The Servo-ON input (SRV-ON) is not connected to COM–.
03	Over-travel inhibition input is valid	P	S	T	F	While Pr5.04 is 0 (Run-inhibition input is valid), • Positive direction over-travel inhibition input (POT) is open and speed command is Positive direction. • Negative direction over-travel inhibition input (NOT) is open and speed command is Negative direction.
04	Torque limit setup is small	P	S	T	F	Either one of the valid torque limit setup value of Pr0.13 (1st) or Pr5.22 (2nd) is set to 5% or lower than the rating.
05	Analog torque limit input is valid.	P	S	T	F	While Pr5.21 is 0 (analog torque limit input accepted), • Positive direction analog torque limit input (P-ATL) is negative voltage and speed command is Positive direction. • Negative direction analog torque limit input (N-ATL) is positive voltage and speed command is Negative direction.
06	INH input is valid.	P	S	T	F	Pr5.18 is 0 (Command pulse inhibition input is valid.), and INH is open.
07	Command pulse input frequency is low.	P	S	T	F	The position command per each control cycle is 1 pulse or smaller due to, • No correct entry of command pulse • No correct connection to the input selected with Pr0.05. • No matching to input status selected with Pr0.06 pr Pr0.07.
08	CL input is valid.	P	S	T	F	While Pr5.17 is 0 (Deviation counter clear at level), the deviation counter clear input (CL) is connected to COM–.
09	ZEROSPD input is valid.	P	S	T	F	While Pr3.15 is 1 (Speed zero clamp is valid.), the speed zero clamp input (ZEROSPD) is open.
10	External speed command is small.	P	S	T	F	While the analog speed command is selected, the analog speed command is smaller than 0.06[V].
11	Internal speed command is 0.	P	S	T	F	While the internal speed command is selected, the internal speed command is set to lower than 30 [r/min]
12	Torque command is small.	P	S	T	F	The analog torque command input (SPR or P-ATL) is smaller than 5 [%] of the rating.
13	Speed limit is small.	P	S	T	F	• While Pr3.17 is 0 (speed is limited by 4th speed of internal speed), Pr3.07, (4th speed of speed setup) is set to lower than 30 [r/min]. • While Pr3.17 is 1 (speed is limited by SPR input), the analog speed limit input (SPR) is smaller than 0.06 [V].
14	Other factor	P	S	T	F	The motor runs at 20 [r/min] or lower even though the factors from 1 to 13 are cleared, (the command is small, the load is heavy, the motor lock or hitting, driver/motor fault etc.)

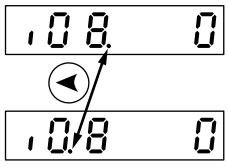
Note * Motor might run even though the other number than 0 is displayed.
Refer to "6.In trouble".

(11) Display of No. of changes in I/O signals



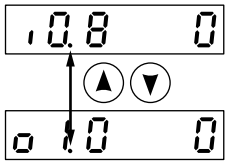
No. of changes in I/O signals (the signal is invalid)
Pin No.
i .....Input signal
o .....Output signal

• Shift the flashing decimal point with ◀.

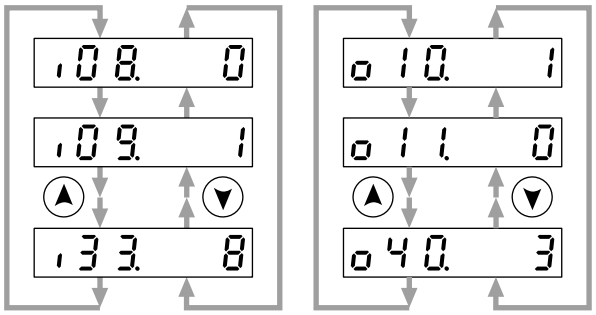


(Left side of decimal point : Pin No. selection)
(Right side of decimal point : Input/Output Pin No. selection)

• The switch of input/output, by pressing ▲ or ▼ button.

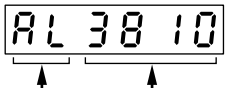


• Select the No. of pin, the number of changes on that pin should be displayed, by pressing ▲ or ▼ button.



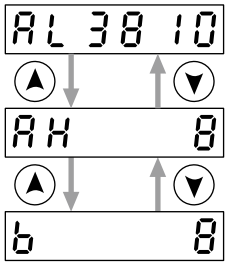
(Lowest place Pin No. of output signal)
(Highest place Pin No. of input signal)

(12) Display of absolute encoder data



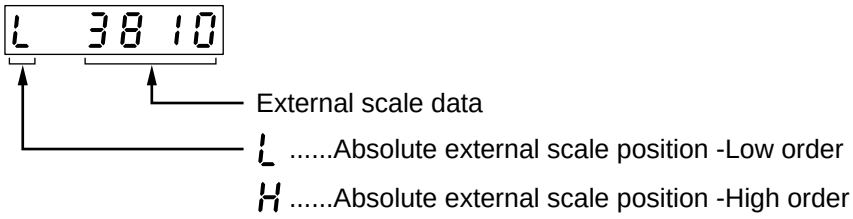
Encoder data
AL .....One revolution data, Low order (L)
AH .....One revolution data, High order (H)
b .....Multi-revolution data

• Select the data to be displayed by pressing ▲ or ▼ button.

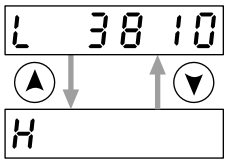


(13) Display of absolute external scale position

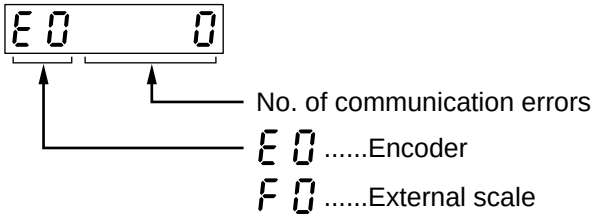
- Displays the absolute position of serial absolute scale.
- If a serial incremental scale, displays the scale position relative to the power on position which is defined as 0.



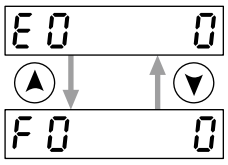
- Select encoder or external scale by pressing ▲ or ▼ button.



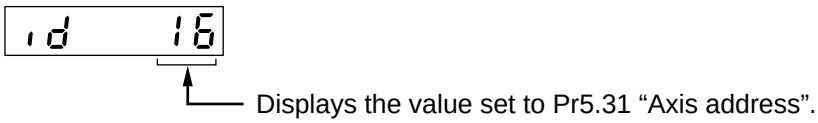
(14) Display of No. of encoder/ external scale communication errors monitor



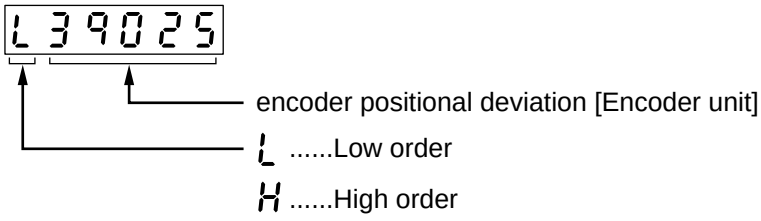
- Select encoder or external scale by pressing ▲ or ▼ button.



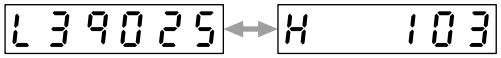
(15) Display of communication axis address



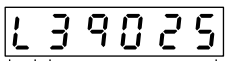
(16) Display of encoder positional deviation [Encoder unit]



- To switch between Low order (L) and High order (H), press ◄.



(17) Display of External Scale Deviation [External Scale Unit]

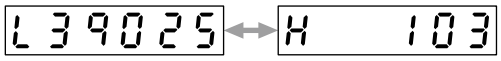


External Scale Deviation [External scale unit]

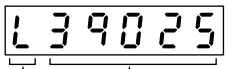
L .....Low order

H .....High order

• To switch between Low order (L) and High order (H), press ◄.



(18) Display of hybrid deviation [Command unit]

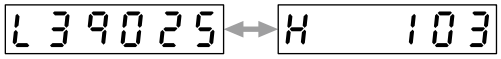


Hybrid deviation [Command unit]

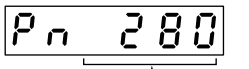
L .....Low order

H .....High order

• To switch between Low order (L) and High order (H), press ◄.

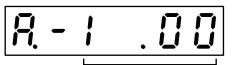


(19) Display of voltage across PN [V]



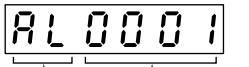
Displays the voltage across PN [V]
(only for reference not an instrument)

(20) Display of Software Version



Displays the software version of the driver.
(Example of display: Ver 1.00)

(21) Display of driver serial number

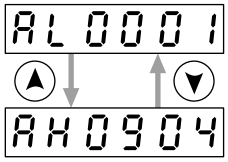


Driver serial number

AL .....Driver serial number- Low order

AH .....Driver serial number- High order

• To switch between Low order (L) and High order (H), press ▲ or ▼.
(Example of display: Serial number 09010001)



(22) Display of motor serial number

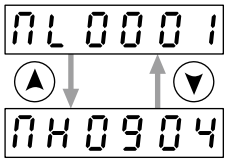


Motor serial number

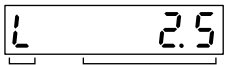
nL .....Motor serial number- Low order

nH .....Motor serial number- High order

• To switch between Low order (L) and High order (H), press ▲ or ▼.
(Example of display: Serial number 09040001)



(23) Display of accumulated operation time

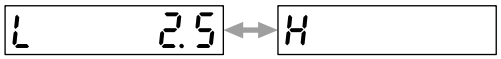


Displays accumulated operation time [h].

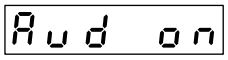
L .....Low order

H .....High order

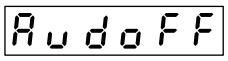
• To switch between Low order (L) and High order (H), press ◄.



(24) Automatic Motor Recognizing Function

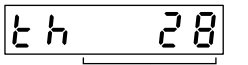


.....Automatic recognition is valid.



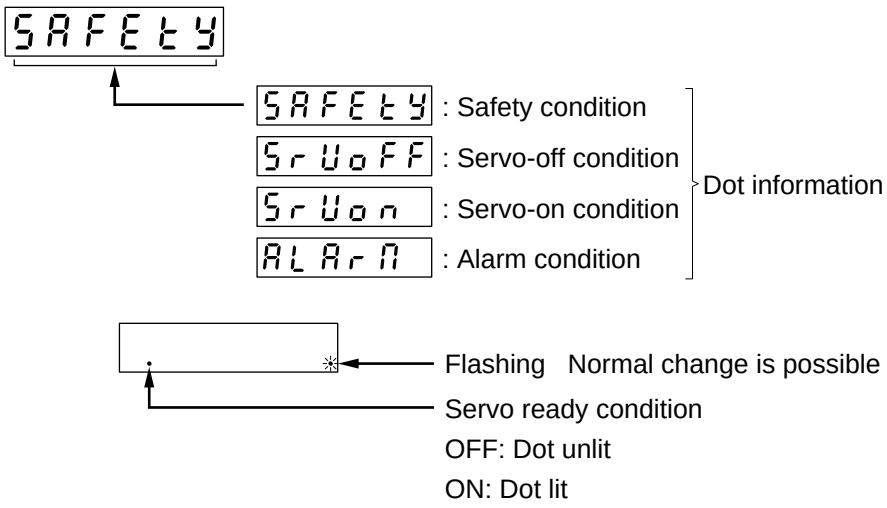
.....Automatic recognition is invalid.

(25) Display of driver temperature

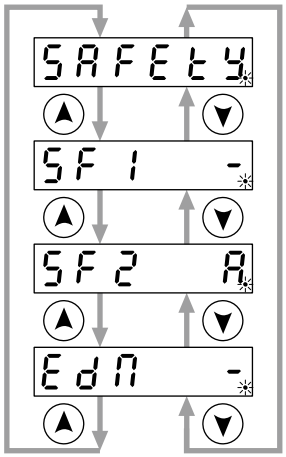


Displays the driver temperature [°C].
(Only for reference not an instrument)

(26) Display of safety condition monitor



• Select desired monitor option by pressing ▲ or ▼ button.



- ...Input photocoupler OFF A ...Input photocoupler ON

- ...Input photocoupler OFF A ...Input photocoupler ON

- ...Output photocoupler OFF A ...Output photocoupler ON

Related page P.7-2 “safety”.

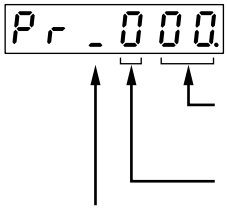
15. How to Use the Front Panel

Parameter Setup Mode

Monitor Mode **SELECTION display**

(M) (Mode switch button)

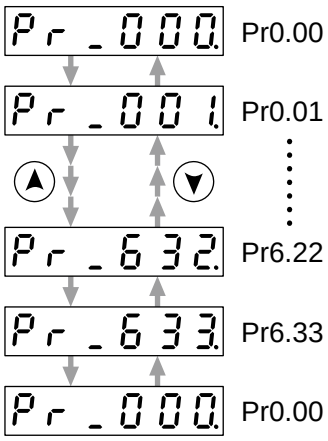
Parameter Setup Mode **SELECTION display**



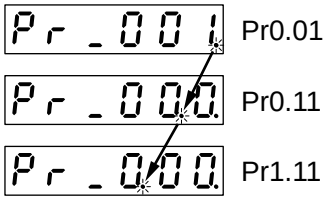
Note

For parameters which place is displayed with " r ", the content changed and written to EEPROM becomes valid after turning off the power once.

- Press ▲ or ▼ to select parameter No. to be set.



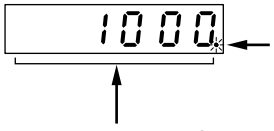
- Press ◄ and the flashing decimal separator shifts to the high order position, allowing the figure at this digit to any other figure.



- Display the No. of parameter to be changed and press (S) to change to EXECUTION display.

(S) (SET button)

EXECUTION display



- Press ▲ or ▼ to set up the value of parameter. (Value increases with ▼ decreases with ▲.)
- Press ◄ and the flashing decimal separator shifts to the high order position, allowing the figure at this digit to any other figure.
- Prolonged depression of (S) updates the parameter in the driver. Note that the parameter value selected by ▲ or ▼ is not reflected until (S) is pressed in this way.
- To cancel the value selected by ▲ or ▼, press (M) instead of (S), and the driver internal parameter value is kept unchanged and the display returns to the parameter number display screen.

Remarks

After changing the parameter value and pressing , the content will be reflected in the control. Do not extremely change the parameter value which change might affect the motor movement very much (especially velocity loop or position loop gains).

(M) (Mode switch button)

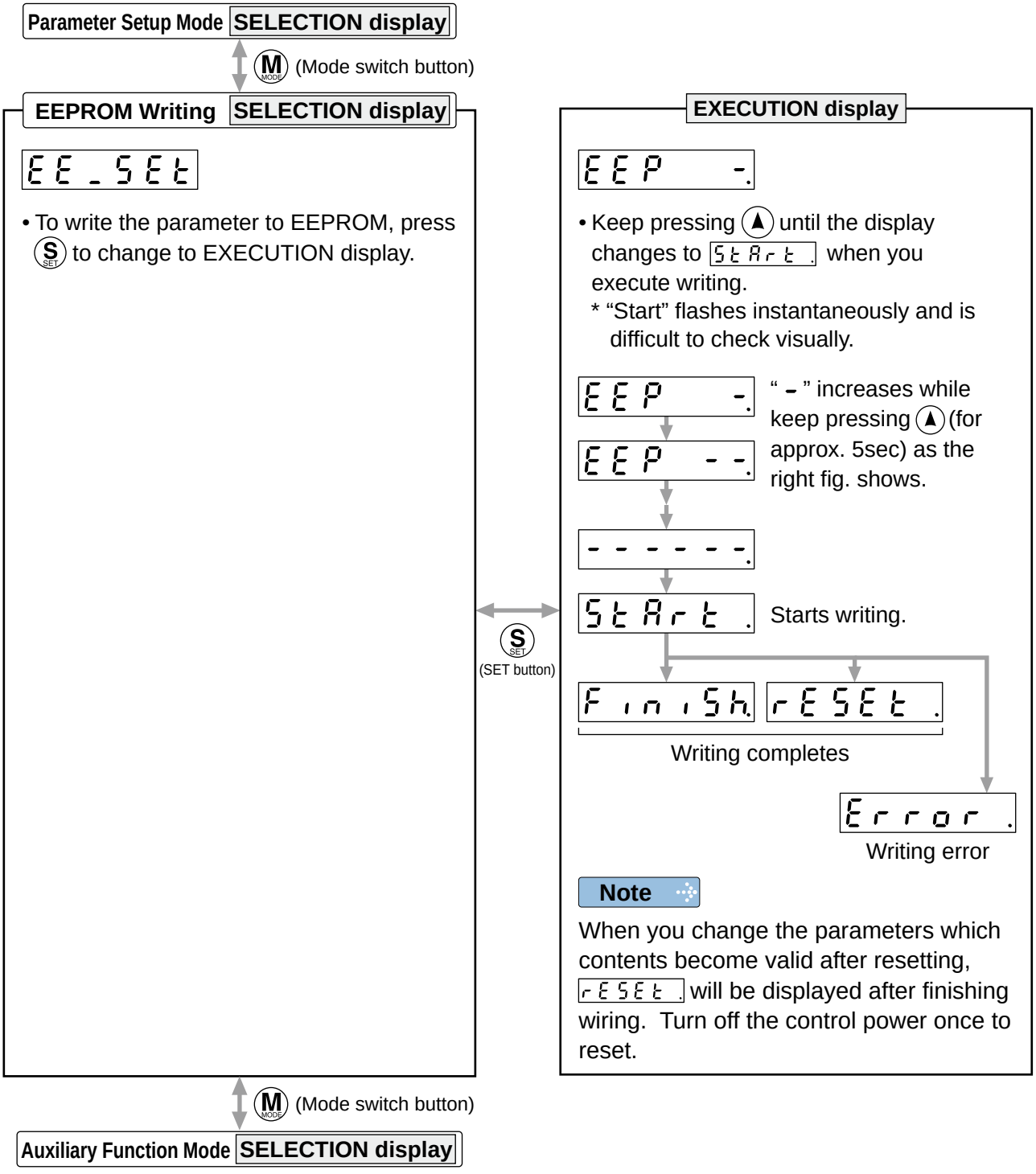
EEPROM Writing **SELECTION display**

Note

- After setting up parameters, return to SELECT mode, referring to structure of each mode (P.2-54).
- Each parameter has a limit in number of places for upper-shifting.

15. How to Use the Front Panel

EEPROM Writing



Note

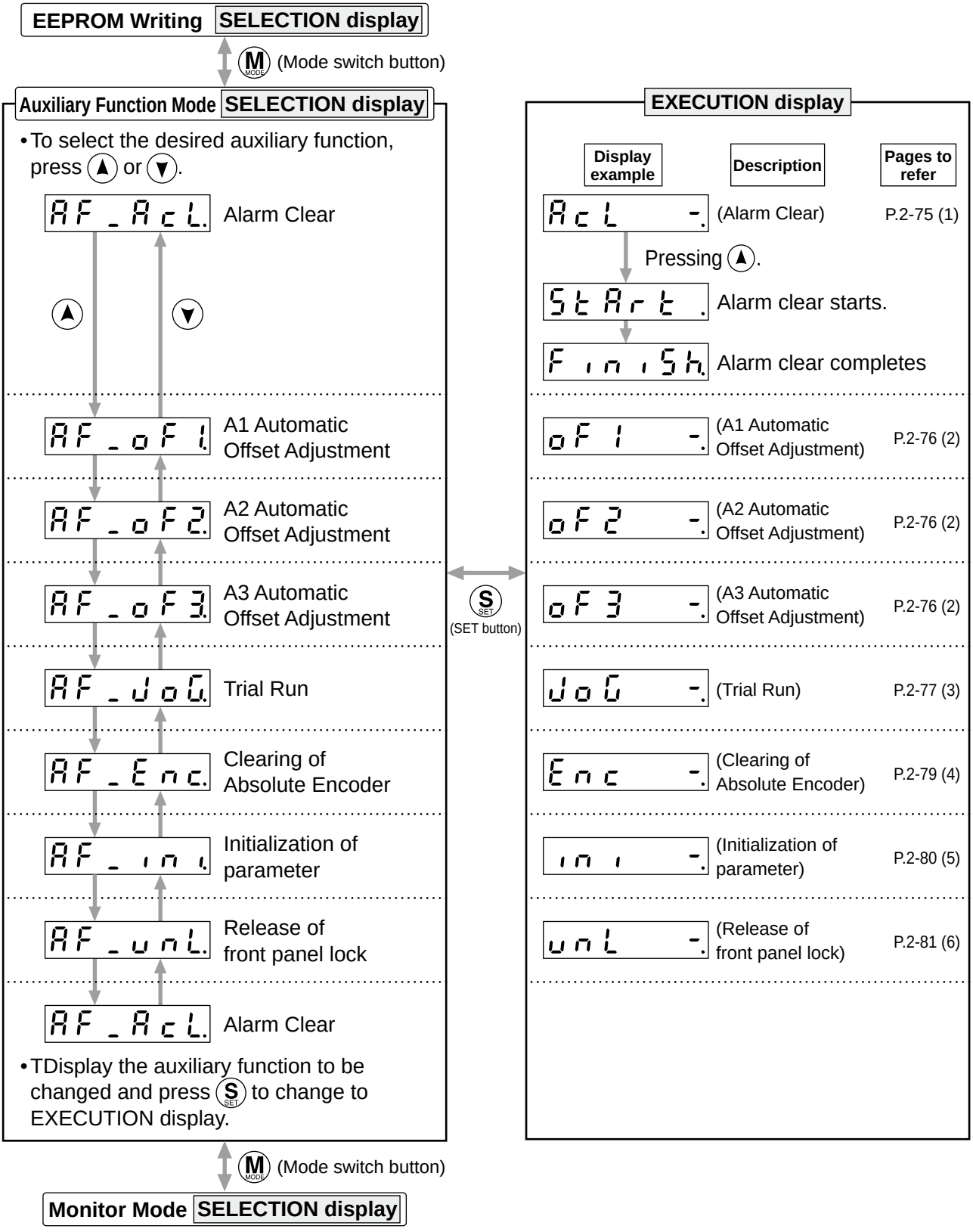
When you change the parameters which contents become valid after resetting, rESEt. will be displayed after finishing wiring. Turn off the control power once to reset.

Caution

1. When writing error occurs, make writing again. If the writing error repeats many times, this might be a failure.
2. Don't turn off the power during EEPROM writing. Incorrect data might be written. If this happens, set up all of parameters again, and re-write after checking the data.
3. When the error defined by Err.11.0 "Under voltage protection of control power supply" occurs, Error. is displayed indicating that no writing is made to EEPROM.

15. How to Use the Front Panel

Auxiliary Function Mode (SELECTION display)

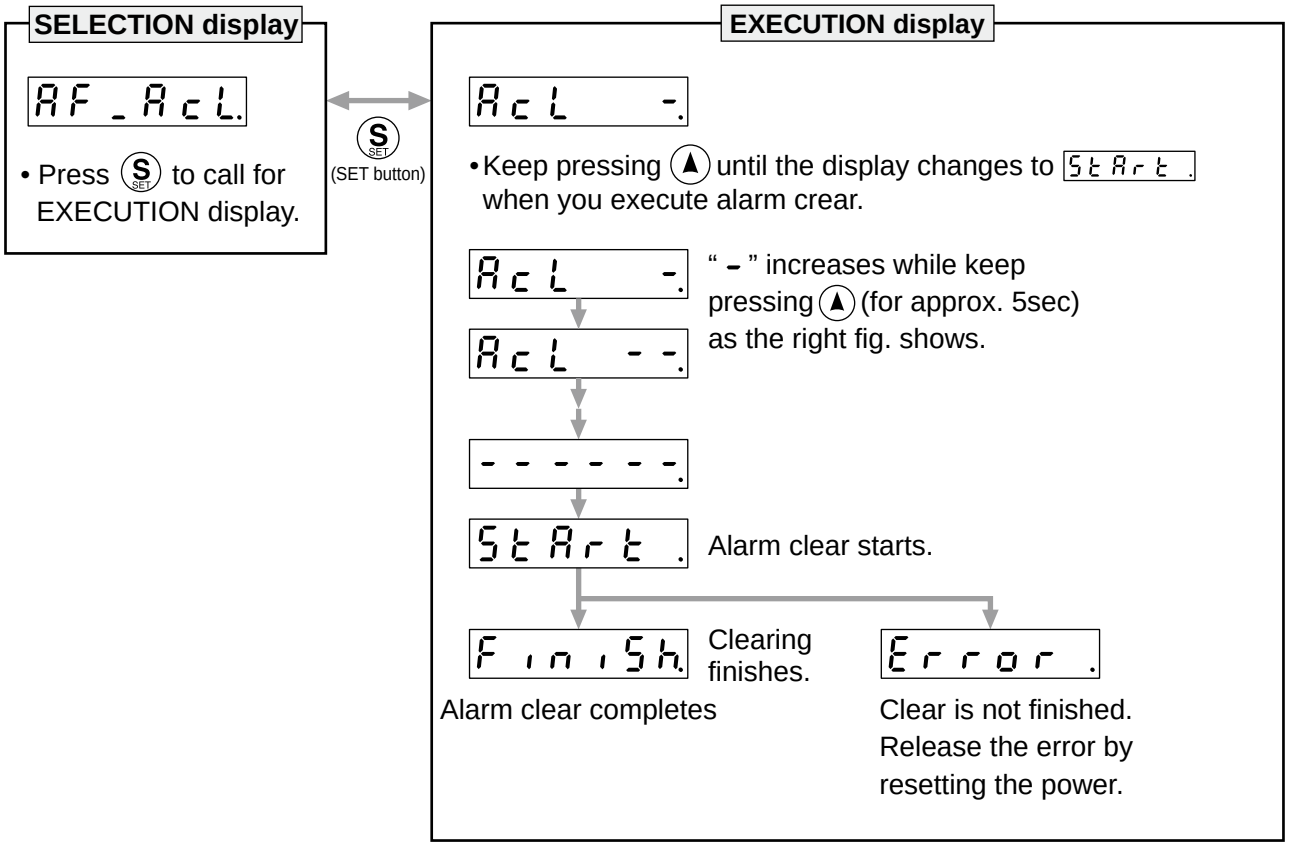


15. How to Use the Front Panel

Auxiliary Function Mode (EXECUTION display)

1) Alarm Clear Screen

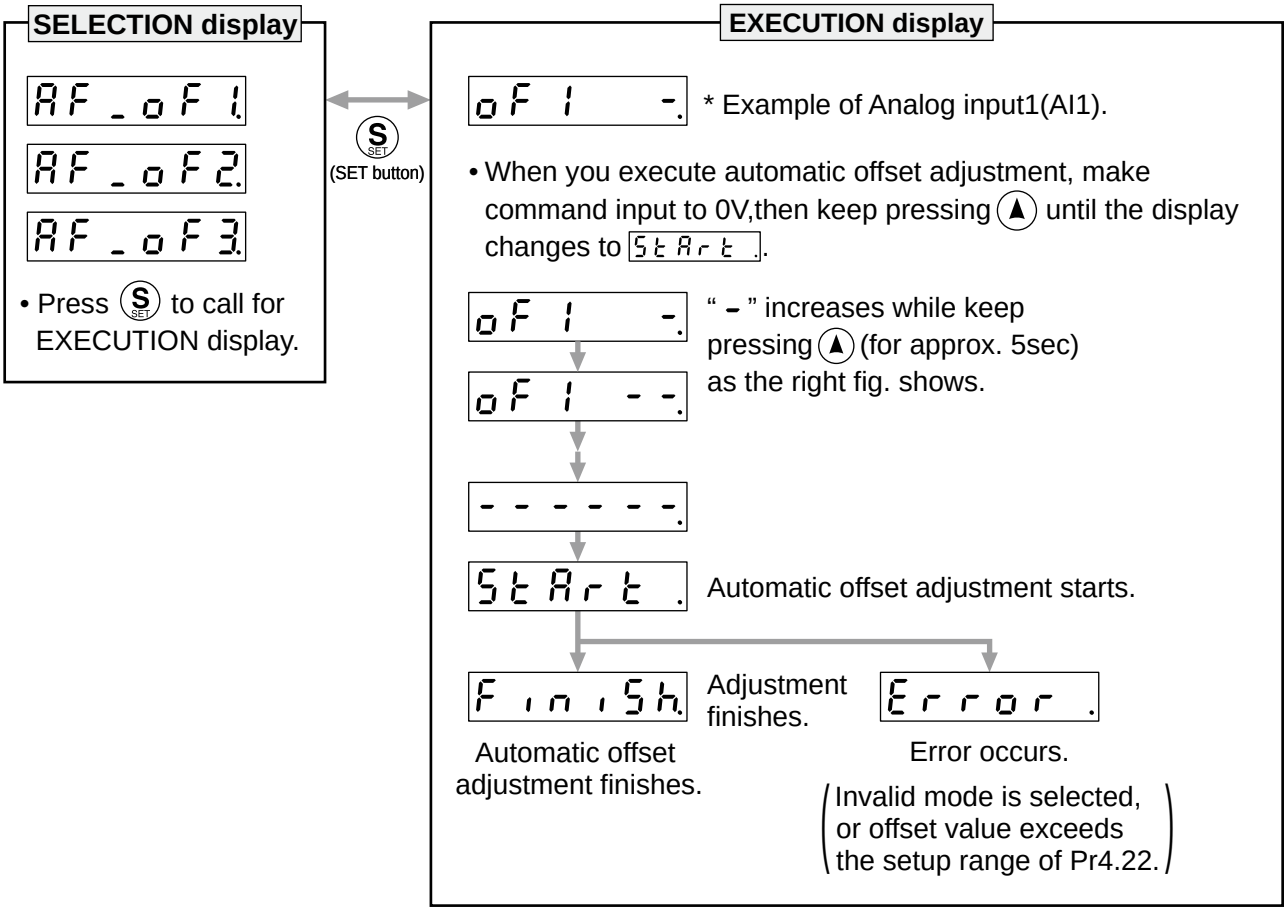
This function releases the current alarm status.
Certain alarms will persist. If this is the case, refer to P.6-3 “When in Trouble - Protective Function”.



Note • After alarm cleaning, return to SELECTION display, referring to structure of each mode (P2-54).

(2) Analog inputs 1 to 3 automatic offset adjustment

This function automatically adjusts offset setting of analog input.
Analog input 1 (AI1).....Pr4.22 (Analog input 1 (AI1) offset setup)
Analog input 2 (AI2).....Pr4.25 (Analog input 2 (AI2) offset setup)
Analog input 3 (AI3).....Pr4.28 (Analog input 1 (AI3) offset setup)



Remarks
- This function is invalid at position control mode.
- You cannot write the data only by executing automatic offset adjustment. Execute a writing to EEPROM when you need to reflect the result afterward.

Note
- After completion of the automatic offset adjustment, return to SELECTION display by referring to P.2-54 “Structure of Each Mode”.

(3) Motor trial run

You can make a trial run (JOG run) without connecting the Connector, Connector X4 to the host controller such as PLC.

Remarks

- Separate the motor from the load, detach the Connector, Connector X4 before the trial run.
- Bring the user parameter setups (especially Pr0.04 and Pr1.01 to 1.04) to defaults, to avoid oscillation or other failure.

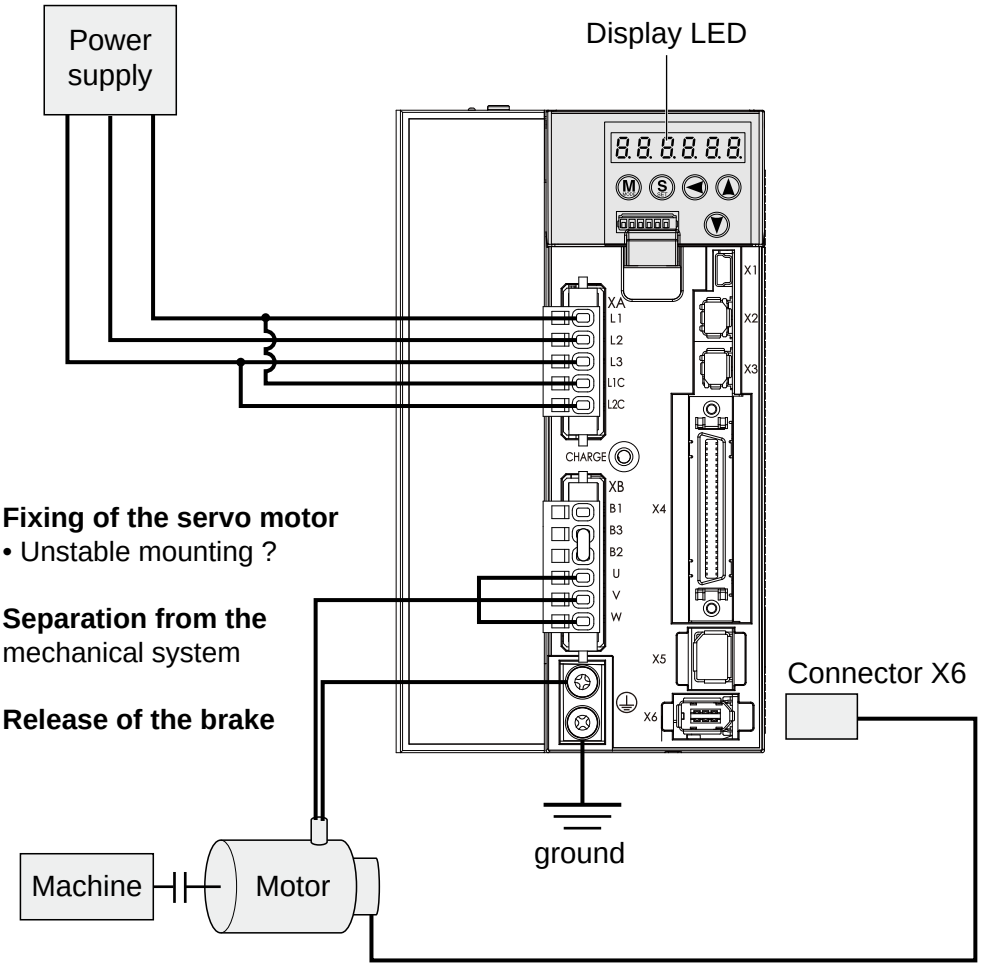
• Inspection Before Trial Run

(1) Inspection on wiring

- Miswiring ? (Especially power input and motor output)
- Short or grounded ?
- Loose connection ?

(2) Confirmation of power supply and voltage

- Rated voltage ?



(3) Fixing of the servo motor

- Unstable mounting ?

(4) Separation from the mechanical system

(5) Release of the brake

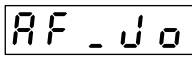
(6) Turn to Servo-OFF after finishing the trial run by pressing (S).

Related page

- For wiring details, refer to P2-9 “Overall Wiring”.
- P.4-6 “Pr0.04”
- P.4-12 “Pr1.01 to 1.04”

• Procedure for Trial Run

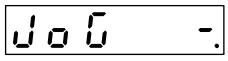
SELECTION display



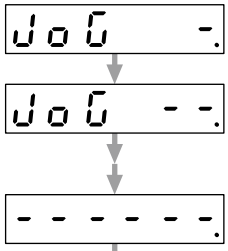
• Press (S SET) to call for EXECUTION display.

(S SET) (SET button)

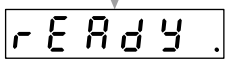
EXECUTION display



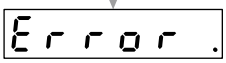
• Keep pressing (▲) until the display changes to rEAdy. when you execute Motor trial run.



" – " increases while keep pressing (▲) (for approx. 5sec) as the right fig. shows.

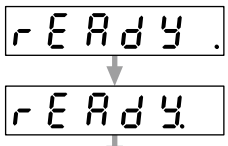


Preparation step 1

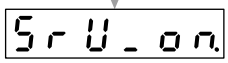


Not a servo-ready status. (Shuts off the main when error occurs.)

• Then keep pressing (◄) until the display of LED changes to SrU_on.



Keep pressing (◄) (approx. 5 sec) to shift the decimal point toward left as the left fig. shows.



Preparation step 2



Not a Servo-Ready. Or SRV-ON signal is not entered.

• **After the Servo-ON of preparation step 2 for trial run,** the motor runs at the preset speed with Pr6.04 (JPG speed) to Positive direction direction by pressing (▲) Negative direction by pressing (▼).

Caution

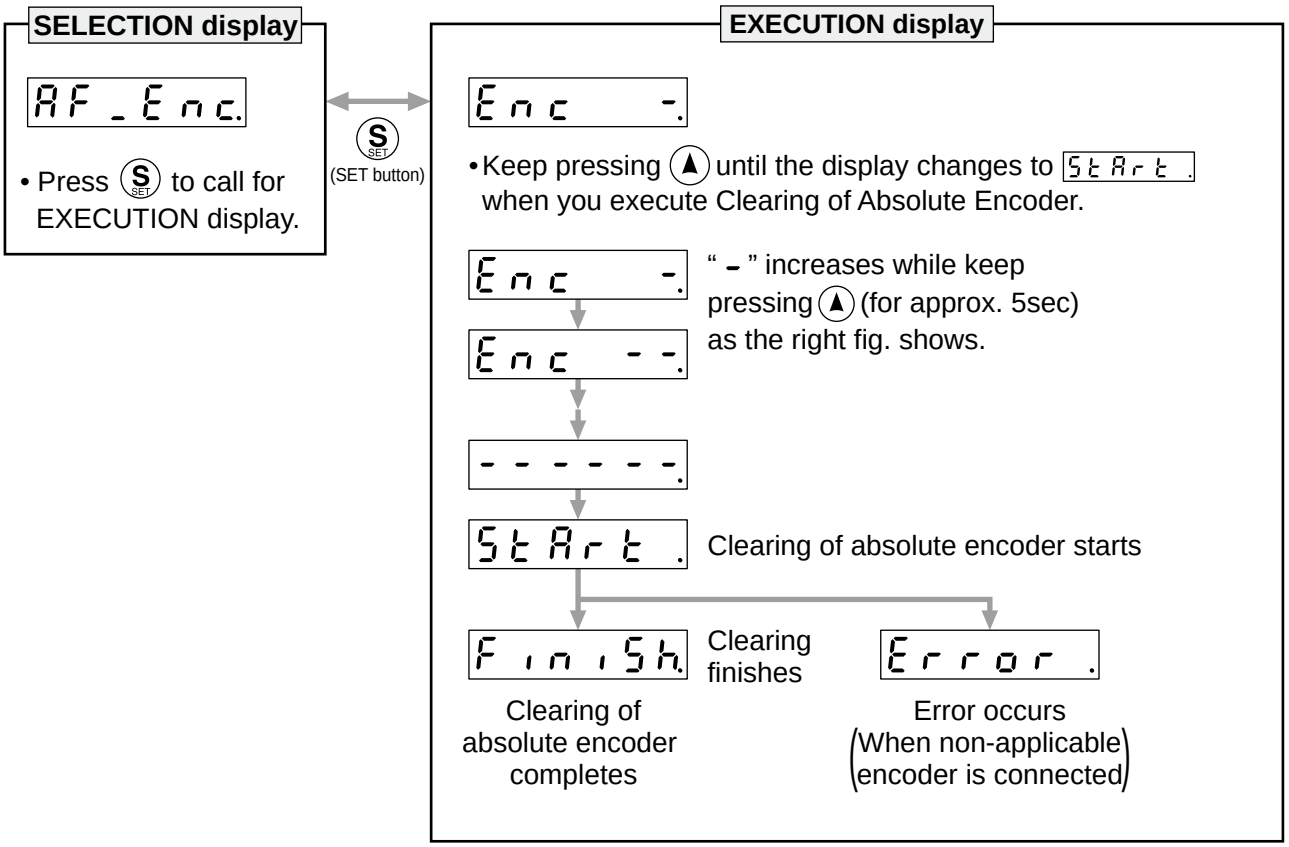
- Before starting the trial run, set the gain-related parameters to appropriate values to avoid problems such as oscillation. If the load is removed, be sure to set Pr0.04 "Inertia Ratio" to 0.
- During the trial run, use the velocity control mode. Various settings including parameters should assure safe and positive operation under appropriate velocity control.
- If SRV-ON becomes valid during trial run, the display changes to Error. which is normal run through external command.
- **If such trouble as disconnection of cable or connector occurs during trial run, the motor makes over-run for maximum 1 sec. Pay an extra attention for securing safety.**

Note

- After finished trial running, return to SELECTION display, referring to structure of each mode (P.2-54).

4) Clearing of Absolute Encoder

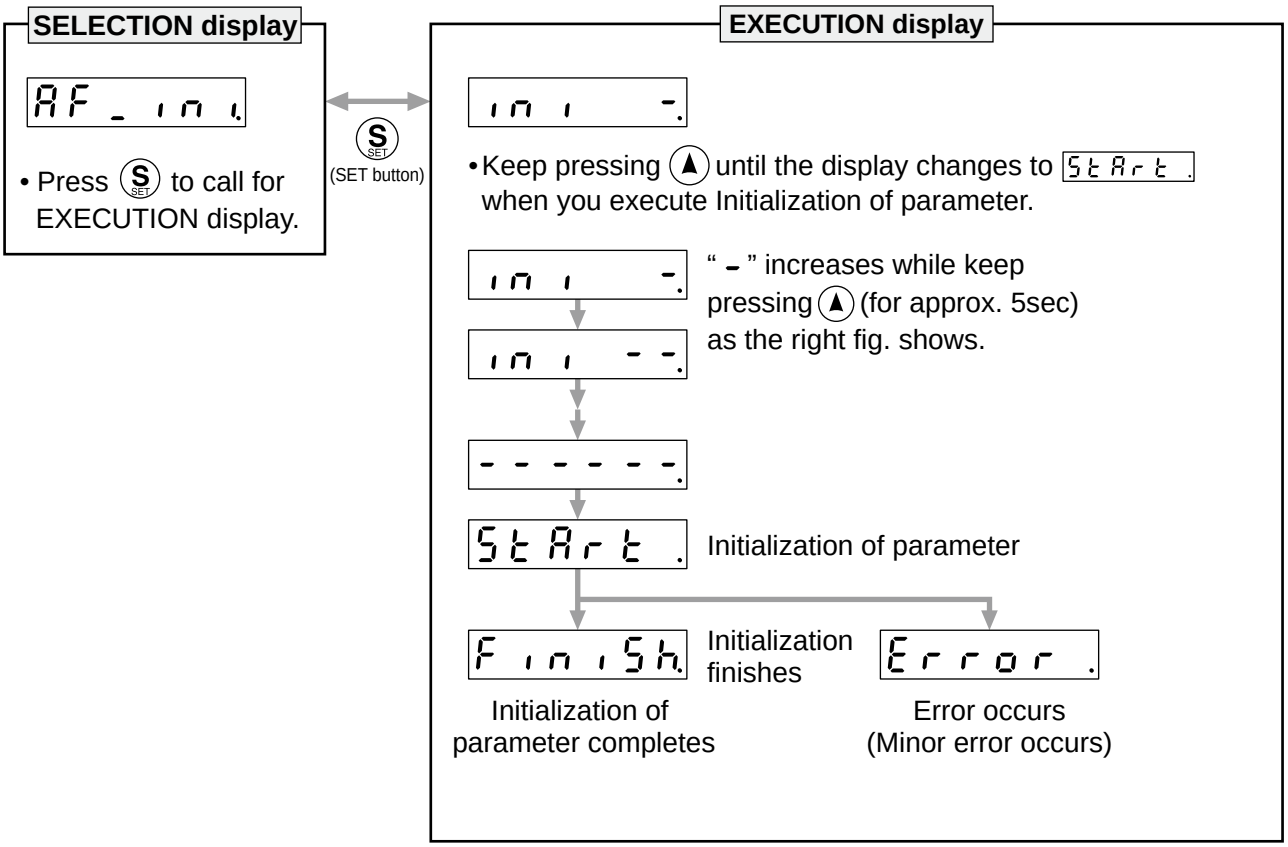
You can clear the multi-turn data of the absolute encoder.



Note • After clearing of absolute encoder finishes, return to SELECTION display, referring to structure of each mode (P.2-54).

(5) Initialization of parameter

Initialize the parameter.

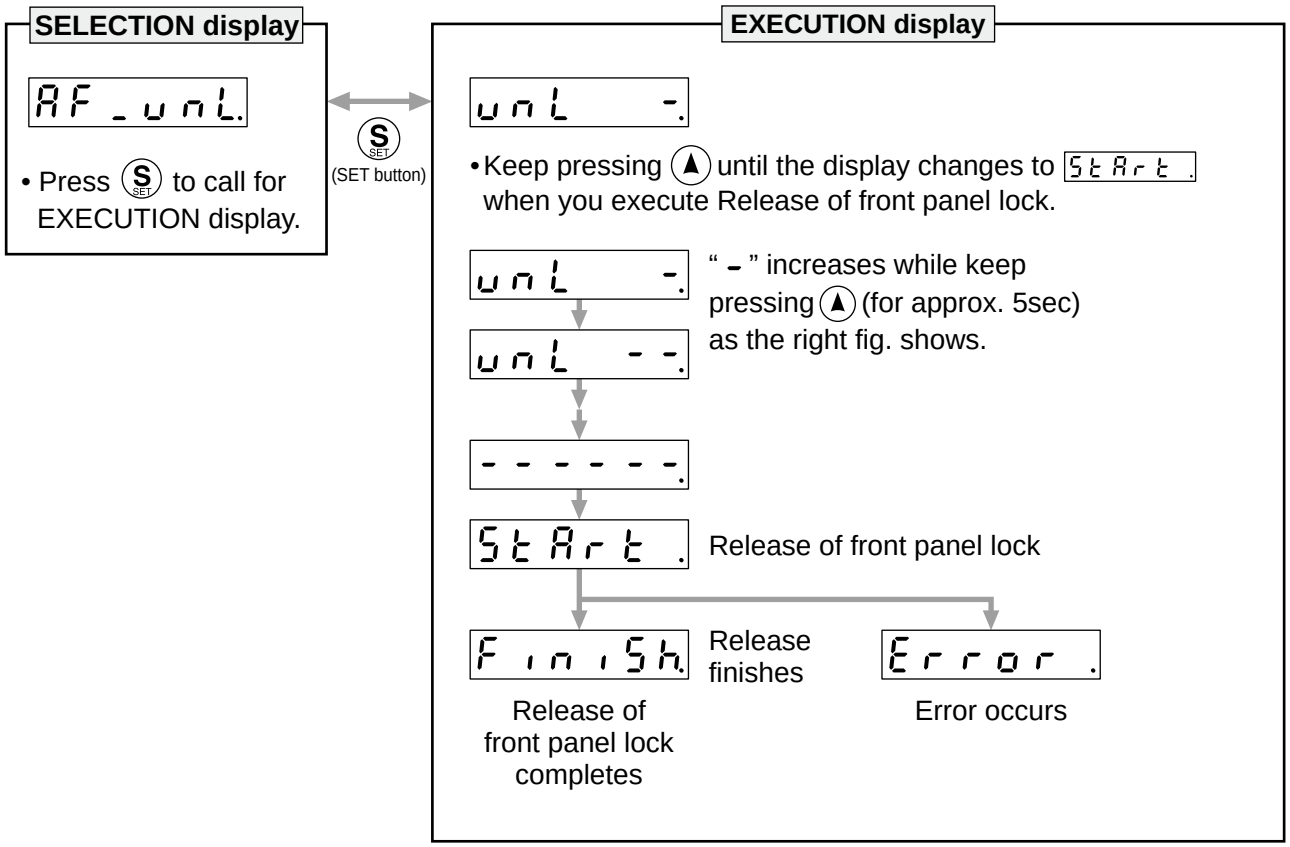


Caution • Parameter cannot be initialized when one of the following error occurs: Err11.0 "Under voltage protection of control power supply", EEPROM related errors (Err36.0, Err36.1, Err36.2, Err37.0, Err37.1 and Err37.2) - initialization will result in "Error" display.

Note • After initialization of parameter finishes, return to SELECTION display, referring to structure of each mode (P.2-54).

(6) Release of front panel lock

Release the front panel lock setting.



Note • After release of front panel lock finishes, return to SELECTION display, referring to structure of each mode (P.2-54).